

Nivell 1

Descàrrega els arxius CSV, estudia'ls i dissenya una base de dades amb un esquema d'estrella que contingui, almenys 4 taules de les quals puguis realitzar les següents consultes:

```
Query 1 sprint4* x
1 • USE sprint4;
2
3 -- N1. Exercici 1. Crea la BBDD amb els CSV proporcionats.
4
5 • SHOW VARIABLES LIKE 'secure_file_priv'; #C:\ProgramData\MySQL\MySQL Server 8.0\Uploads
6 # carpeta segura para importar CSVs to WB
7

Query 1 sprint4* x
10 #Primer crearem les taules a MySQL, a continuació li introduïm les dades dels CSV i finalment creem les constraints necessàries per
11 -- crear les FK i enllassar les taules.
12 • CREATE TABLE IF NOT EXISTS transactions (
13     id VARCHAR(100) PRIMARY KEY,
14     card_id VARCHAR(100),
15     company_id VARCHAR(100),
16     timestamp timestamp,
17     amount DECIMAL(10, 2),
18     declined TINYINT(1),
19     products_ids TEXT,
20     user_id VARCHAR(100),
21     lat FLOAT,
22     longitude FLOAT
23 );
24

Query 1 sprint4* x
25 • LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/transactions.csv'
26 INTO TABLE transactions
27 FIELDS TERMINATED BY ';'
28 LINES TERMINATED BY '\n'
29 IGNORE 1 LINES;
30
31 • SELECT*
32 FROM transactions; #Comprovem que la taula s'hagi creat correctament.
33
34 • ALTER TABLE transactions #Aqui debo modificar el tipo de variable para que coincida con data_user.id y poder enlazar las tablas (FK).
35 MODIFY user_id INT;
36

Query 1 sprint4* x
38 • CREATE TABLE IF NOT EXISTS companies (
39     id VARCHAR(100) PRIMARY KEY,
40     company_name VARCHAR(100),
41     phone VARCHAR(100),
42     email VARCHAR(100),
43     country VARCHAR(100),
44     website VARCHAR(100)
45 );
46
47 • LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/companies.csv'
48 INTO TABLE companies
49 FIELDS TERMINATED BY ','
50 LINES TERMINATED BY '\n'
51 IGNORE 1 LINES;
```

```
Query 1 sprint4* x
CREATE TABLE IF NOT EXISTS credit_cards (
    id VARCHAR(100) PRIMARY KEY,
    user_id VARCHAR(100),
    iban VARCHAR(100),
    pan VARCHAR(100),
    pin VARCHAR(100),
    cvv VARCHAR(100),
    track1 VARCHAR(100),
    track2 VARCHAR(100),
    expiring_date VARCHAR(100)
);
```

```
Query 1 sprint4* x
LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/credit_cards.csv'
INTO TABLE credit_cards
FIELDS TERMINATED BY ','
LINES TERMINATED BY '\n'
IGNORE 1 LINES;

SELECT*
FROM credit_cards;
```

```
Query 1 sprint4* x
CREATE TABLE IF NOT EXISTS data_users (
    id INT PRIMARY KEY,
    name VARCHAR(100),
    surname VARCHAR(100),
    phone VARCHAR(100),
    email VARCHAR(100),
    birth_date VARCHAR(100),
    country VARCHAR(100),
    city VARCHAR(100),
    postal_code VARCHAR(100),
    address VARCHAR(300)
);

LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/data_users.csv'
INTO TABLE data_users
FIELDS TERMINATED BY ','
OPTIONALLY ENCLOSED BY '"'
ESCAPED BY '\\'
LINES TERMINATED BY '\r\n'
IGNORE 1 LINES;

SELECT*
FROM data_users;
```

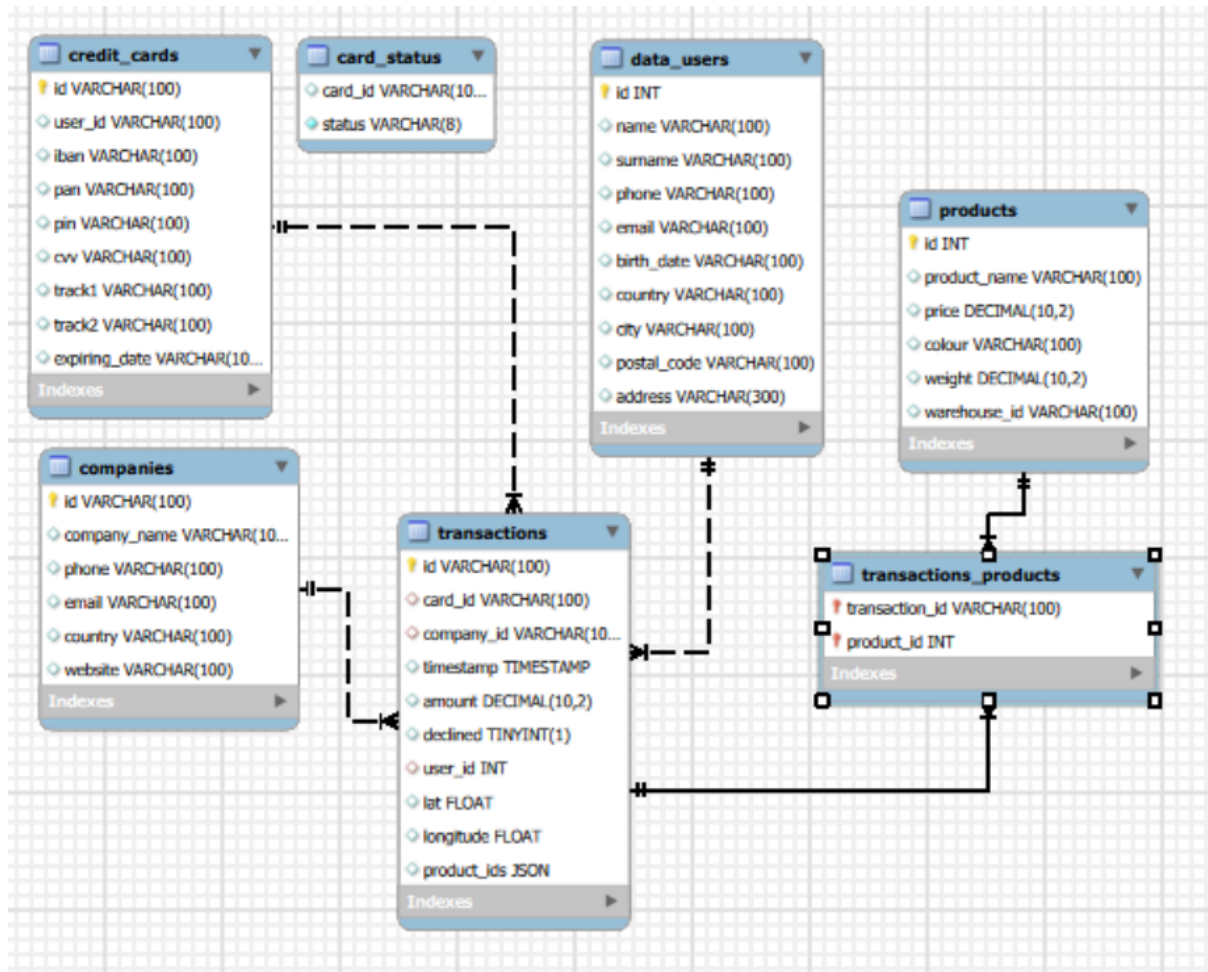
```
Query 1 sprint4* x
-- Creamos las FK para enlazar las tablas -----

ALTER TABLE transactions
ADD CONSTRAINT fk_transactions_companies
FOREIGN KEY (company_id)
REFERENCES companies(id);

ALTER TABLE transactions
ADD CONSTRAINT fk_transactions_card
FOREIGN KEY (card_id)
REFERENCES credit_cards(id);

ALTER TABLE transactions
ADD CONSTRAINT fk_transactions_user
FOREIGN KEY (user_id)
REFERENCES data_users(id);
```

DIAGRAMA FINAL



Exercici 1

Realitza una subconsulta que mostri tots els usuaris amb més de 80 transaccions utilitzant almenys 2 taules.

```
127 • SELECT *
128 FROM data_users du
129 INNER JOIN (
130     SELECT du.id, COUNT(t.id) as suma_transaccions_usuari
131     FROM transactions t
132     INNER JOIN data_users du
133     ON t.user_id = du.id
134     WHERE t.declined = 0 #Interpretamos que no se contabilizan las transacciones declinadas.
135     GROUP BY du.id
136     HAVING suma_transaccions_usuari > 80
137 ) TT
138 ON du.id = TT.id;
139
```

id	name	surname	phone	email	birth_date	country	city	postal_code	address	id	suma_transaccions_usuari
318	Bnyr	Astuw	+33-120-9644	bnyr.astuw@example.com	May 3, 1974	Italy	Genoa	16100	53 Astuw Street	318	86
289	Dxxvgi	Hwrcu	+98-309-8797	dxvgi.hwrcu@example.com	Aug 20, 1976	Germany	Stuttgart	70173	82 Hwrcu Street	289	91
185	Molly	Gilliam	0800 120 8023	donec@outlook.couk	Dec 21, 1993	United Kingdom	London	EC1A 1BB	P.O. Box 202, 5638 Mi Rd.	185	107

Result 12 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
12	12:16:25	SELECT * FROM data_users du INNER JOIN (SELECT du.id, COUNT(t.id) as suma...	3 row(s) returned	0.641 sec / 0.000 sec

Exercici 2

Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2 taules.

```
143 • SELECT cc.iban, ROUND(AVG(t.amount), 2) AS 'avg_amount_x_iban'
144 FROM transactions t
145 INNER JOIN credit_cards cc ON t.card_id = cc.id
146 INNER JOIN companies c ON t.company_id = c.id
147 WHERE c.company_name = 'Donec Ltd'
148 GROUP BY cc.iban;
```

iban	avg_amount_x_iban
XX911406401125586307586805	356.25
SK9446370242474562577506	142.96
XX776752917845952975555640	257.37
XX413827362289719304908990	139.59
XX347787246070769610780308	240.41

Result 18 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
18	12:18:13	SELECT cc.iban, ROUND(AVG(t.amount), 2) AS 'avg_amount_x_iban' FROM transa...	371 row(s) returned	0.031 sec / 0.000 sec

Nivell 2

Crea una nova taula que reflecteixi l'estat de les targetes de crèdit basat en si les tres últimes transaccions han estat declinades aleshores és inactiu, si almenys una no és rebutjada aleshores és actiu. Partint d'aquesta taula respon:

Exercici 1

Quantes targetes estan actives?

```
153 • CREATE TABLE card_status AS
154 SELECT
155     card_id,
156     CASE
157         WHEN MIN(declined) = 1 THEN 'inactiva'
158         ELSE 'activa'
159     END AS status
160 FROM (
161     SELECT t.card_id, t.declined, t.timestamp, ROW_NUMBER() OVER (
162         PARTITION BY t.card_id ORDER BY timestamp DESC) AS rn FROM transactions t) TT
163 WHERE rn <= 3
164 GROUP BY card_id;
```

Query 1 **sprint4***

-- Quantes targetes estan actives?

```
171 • SELECT COUNT(cs.card_id) AS targetes_actives
172 FROM card_status cs
173 WHERE status = 'activa'
174 GROUP BY cs.status;
```

Result Grid

targetes_actives
4995

Result 23

Output

Action Output

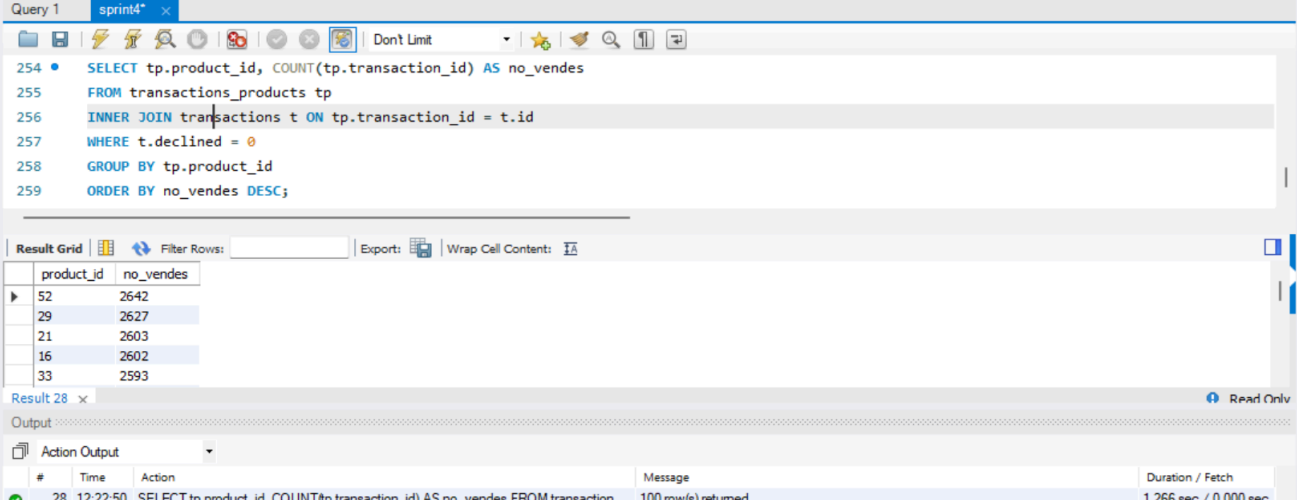
#	Time	Action	Message	Duration / Fetch
23	12:20:55	SELECT COUNT(cs.card_id) AS targetes_actives FROM card_status cs WHERE st...	1 row(s) returned	0.015 sec / 0.000 sec

Nivell 3

Exercici 1

Crea una taula amb la qual puguem unir les dades del nou arxiu products.csv amb la base de dades creada, tenint en compte que des de transaction tens product_ids. Genera la següent consulta:

Necessitem conèixer el nombre de vegades que s'ha venut cada producte.



The screenshot shows a SQL query editor with a query window and a results window.

Query 1 (sprint4*)

```
254 • SELECT tp.product_id, COUNT(tp.transaction_id) AS no_vendes
255 FROM transactions_products tp
256 INNER JOIN transactions t ON tp.transaction_id = t.id
257 WHERE t.declined = 0
258 GROUP BY tp.product_id
259 ORDER BY no_vendes DESC;
```

Result Grid

	product_id	no_vendes
▶	52	2642
	29	2627
	21	2603
	16	2602
	33	2593

Result 28 (Read Only)

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 28	12:22:50	SELECT tp product_id, COUNT(tp.transaction_id) AS no_vendes FROM transaction...	100 row(s) returned	1.266 sec / 0.000 sec